

Optipeople DemoCNC D-2800M — Error & Alarm Code Reference

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Codes are namespaced by subsystem. Format: <S>-<NNNN> where S is the severity letter and NNNN is the four-digit identifier.

Letter	Severity	Effect
W	Warning	Machine continues; HMI shows yellow badge
E	Error	Cycle aborted; machine stops in a safe state; can be acknowledged after fixing
F	Fault	Power-cycle required; usually a hardware failure
S	Safety	Safety-rated event; logged immutably; engineer sign-off may be required

Number ranges:

Range	Subsystem
1000–1999	Motion / axis
2000–2999	Safety / interlocks
3000–3999	Spindles
4000–4999	Tool change / magazine
5000–5999	Pneumatic / vacuum
6000–6999	Dust extraction
7000–7999	Control / HMI / IPC
8000–8999	Network / integration
9000–9999	Power / cabinet

Motion / axis (1000–1999)

W-1001 — Soft limit approaching

Cause: Commanded position is within 5 mm of a soft limit. **Recover:** Reduce travel or recheck job offsets. The cycle continues unless the limit is reached, which escalates to E-1101.

W-1010 — Following error rising

Cause: Servo following error trending up over the last 30 s, still below the abort threshold. **Recover:** Often the first sign of insufficient lubrication. Run the manual grease procedure (maintenance §3) for the affected axis. If it recurs, inspect the axis for binding.

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E-1101 — Soft limit reached, axis stopped

Cause: Axis hit a software-defined travel limit. **Recover:** Switch to MAN mode, jog away from the limit, re-home, re-run the job. Check job for incorrect zero offset.

E-1102 — Hard limit reached

Cause: Physical limit switch tripped. Should not happen in normal use — implies homing was lost or a mechanical issue. **Recover:** Power cycle, re-home. If recurs, the limit switch may be faulty or the home reference has drifted.

E-1201 — Following error abort (X)

E-1202 — Following error abort (Y)

E-1203 — Following error abort (Z)

Cause: Servo could not follow the commanded trajectory. Likely causes (in order of frequency): collision, dull tool, insufficient lube, drive fault. **Recover:**

1. Inspect for collision (panel shifted, tool broken).
2. Check the tool's wear state.
3. Run manual grease per maintenance §3.
4. Check drive diagnostics — see §7 below.

E-1301 — Homing failed (X)

E-1302 — Homing failed (Y)

E-1303 — Homing failed (Z)

Cause: Home sensor not seen during the homing sweep. **Recover:** Verify no obstruction. Check the home prox sensor LED — it should briefly light as the axis crosses the dog. If sensor is dead, replace (PN: OP-PRX-M12).

F-1901 — Servo drive comm lost (X / Y / Z)

Cause: EtherCAT slave on the axis drive is offline. **Recover:** Power cycle. If recurs, check the EtherCAT trunk LEDs on the drive (top: link OK; bottom: pulses during traffic). If the bottom LED is dark, replace the drive's EtherCAT cable.

Safety / interlocks (2000–2999)

E-2010 — Door open during cycle

Cause: A safety door was opened while in AUTO mode. **Recover:** Close the door. Acknowledge the alarm. The cycle does not auto-resume — operator must press CYCLE START again.

E-2020 — Light curtain blocked

Cause: Optional finger-class light curtain triggered. **Recover:** Clear the protected zone. Acknowledge. The HMI logs the event with which beam(s) were blocked.

E-2030 — Safety relay mismatch

Cause: The two channels of a safety circuit (E-stop or door) read different states for > 200 ms. **Recover:** Inspect for damaged switch or cable. Power cycle. If recurs, **Service** — the switch may be failing.

S-2090 — Override attempt detected

Cause: A door switch was bypassed or a key was left in the override position. **Recover:** Acknowledgment requires supervisor PIN. The event is written to the immutable safety log and reviewed at audit.

S-9001 — Emergency stop active (also under power)

Cause: An E-stop button is pressed. **Recover:** Release the button (twist or pull, per type). Reset on the HMI. CYCLE START requires fresh press.

Spindles (3000–3999)

W-3010 — Spindle temp high (S1..S6)

Cause: PT100 reading exceeds 70 °C (warning threshold). **Recover:** Reduce duty / feed. If sustained for 5 minutes, escalates to `E-3110`.

W-3020 — Spindle vibration above baseline

Cause: Vibration measurement on the active spindle drifted > 30 % above its 30-day baseline. **Recover:** Tool may be unbalanced (chipped insert, gummed-up flutes) or holder seated incorrectly. Replace / reseal the tool.

E-3110 — Spindle overheated (S1..S6)

Cause: PT100 exceeded 85 °C. **Recover:** Stop and let cool to < 50 °C before restarting. Inspect cooling air paths (dust on hood, blocked exhaust). If recurs at light load, spindle bearing degradation — schedule \$9 TBO.

E-3210 — VFD overcurrent (S1..S6)

Cause: Drive tripped on overcurrent. Often a stalled tool or chip welded to the flute. **Recover:** Stop, inspect the tool, check for stuck part. Acknowledge and restart from the failed operation.

E-3220 — VFD bus undervoltage

Cause: Mains voltage dropped or VFD bus cap is failing. **Recover:** Check building voltage. If voltage is normal, **Service**.

E-3310 — Spindle did not reach commanded speed

Cause: Spindle ramped to < 90 % of commanded RPM within 10 s. **Recover:** Probable belt slip (S3) or motor coupling fault. Inspect belt tension (maintenance \$5.4) for S3. For other spindles: **Service**.

F-3901 — Spindle encoder lost

Cause: Spindle encoder feedback flatlined. **Recover:** Power cycle. If recurs, encoder cable or encoder board failure — **Service**.

Tool change / magazine (4000–4999)

W-4001 — Tool life expired

Cause: Spindle hours on the loaded tool exceeded its configured life. **Recover:** Replace the tool. Reset the tool's life counter in **Tools** → **Edit** → **Reset Life**.

E-4101 — ATC pickup failed

Cause: Vertical spindle did not register the tool present after pickup attempt. **Recover:** Inspect the slot — chip in the taper, missing pull stud, or holder loaded backwards. Manually purge in **Tools** → **Magazine** → **Manual Purge** then retry.

E-4102 — ATC return failed

Cause: Magazine slot did not register the tool home after return. **Recover:** Open the magazine guard, verify the tool is correctly seated. If holder is jammed, follow the manual extraction procedure (maintenance app note D2800M-AN-04).

E-4110 — ATC clamp/unclamp solenoid fault

Cause: The pull-stud solenoid did not change state in time. **Recover:** Check air pressure (≥ 5.5 bar). If air OK, **Service**.

E-4201 — Tool offset out of range

Cause: Calibrated tool length differs from previous value by > 2 mm, or measured diameter differs by > 0.5 mm. **Recover:** Tool may be broken or wrong tool loaded. Inspect and recalibrate.

Pneumatic / vacuum (5000–5999)

W-5001 — Air pressure low

Cause: PS1 reading < 5.0 bar. **Recover:** Check the building compressor, FRL bowl drain (water accumulation), and FRL filter elements.

E-5101 — Air pressure critical

Cause: PS1 < 4.5 bar. **Recover:** Machine refuses to start a cycle until pressure is restored. Find and fix the air supply issue. Common: shared compressor at peak demand, leaking presser cylinder.

E-5201 — Presser cylinder not extended (1..6)

Cause: PS2..PS7 sensor did not confirm extended state within 1.5 s. **Recover:** Inspect the cylinder, the proximity sensor (LED on body should light at extend), and the air line. The most common cause is a panel that doesn't allow the presser to seat — wrong panel size or warped material.

E-5301 — Vacuum hold-down weak

Cause: Vacuum sensor on the active zone read pressure > -300 mbar (weaker than spec) for 3 s during a cut move. **Recover:** Stop the cycle. Brush gasket clean, look for tears, confirm correct zones are selected for the

panel size.

E-5310 — Vacuum lost during cycle

Cause: Active zone reads atmospheric pressure. **Recover:** Stop immediately. Likely a torn gasket strip — replace per maintenance §6.2.

Dust extraction (6000–6999)

W-6001 — Dust extractor flow low

Cause: Flow sensor reading below 4,500 m³/h (warning). **Recover:** Check that the extractor is on. Inspect filter cake on the extractor — likely needs cleaning or replacement.

E-6101 — Dust extractor offline

Cause: Dust extractor not detected. **Recover:** Machine refuses to start a cycle. Turn on the extractor. Confirm the auxiliary contact wiring at TB-EXT is intact.

Control / HMI / IPC (7000–7999)

W-7001 — IPC disk space low

Cause: OS drive has < 15 % free. **Recover:** Archive old job files from **Jobs** → **Archive**. The HMI auto-cleans `tmp` if free space falls below 5 %.

W-7010 — IPC clock drift

Cause: RTC drifted > 60 s. Will affect job timestamps and CMMS sync. **Recover:** Sync time at **Settings** → **System** → **Time**. If recurs every power-up, replace the IPC RTC battery (PN: OP-IPC-BAT).

E-7110 — Motion controller heartbeat lost

Cause: HMI lost contact with the OptiMotion controller. **Recover:** Power cycle. If recurs, check the internal Ethernet patch between IPC and motion controller, and the motion controller's status LEDs.

F-7901 — Configuration corrupted

Cause: `machine.cfg` failed checksum on boot. **Recover:** HMI auto-restores from the daily backup (in `C:\Optipeople\config\backup\`). If the backup also fails, **Service** — config restore from USB.

Network / integration (8000–8999)

W-8001 — CMMS sync failed

Cause: Last scheduled upload to the configured CMMS endpoint returned an error. **Recover:** Check the URL & API token at **Settings** → **Integration** → **CMMS**. Manual retry via **Test Connection**.

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W-8010 — Remote support tunnel offline

Cause: TeamViewer Tensor connection is down. **Recover:** Check internet connectivity from the IPC. If LAN OK, restart the Tensor service from **Settings** → **Remote Support**.

E-8101 — Job import rejected (format)

Cause: Uploaded MPR / DXF / CSV failed parse. **Recover:** Open the file in the source CAD/CAM tool and re-export. Common causes: text in numeric fields, missing units header, MPR revision mismatch.

Power / cabinet (9000–9999)

W-9001 — Cabinet temperature high

Cause: Internal cabinet > 45 °C. **Recover:** Check the intake filter and the cabinet fan. Ambient over 35 °C reduces VFD headroom — consider improving shop ventilation.

E-9101 — Phase loss

Cause: Phase monitor K1 detected loss or unbalance on one of L1/L2/L3. **Recover:** Investigate building supply. Do not attempt to bypass the phase monitor.

E-9102 — Phase reversal

Cause: Phase rotation reversed (e.g. after a power infrastructure change). **Recover:** Qualified electrician swaps two phases at the supply disconnect.

F-9901 — 24 V DC rail fault

Cause: 24 V PSU has shut down (over-current or under-voltage). **Recover:** Power cycle. If recurs, identify and disconnect the load that's drawing too much (often a shorted I/O cable).

Diagnostic tips

1. **Note the exact code and timestamp.** The HMI log under **Status** → **Alarm Log** keeps the last 5,000 events; export with **Export Log** before clearing.
2. **Look at what happened just before.** Many faults are downstream effects of an earlier warning — e.g. W-1010 precedes E-1201 by minutes or hours.
3. **Reproduce intentionally.** If a fault only occurs during a specific job step, isolate that step in MAN / SETUP mode.
4. **Open the support ticket with logs attached.** From **Settings** → **Support**, the "Create Ticket" button bundles the latest logs and snapshots into a single archive.

Support contact

- support@optipeople.dk
- +45 70 70 70 70 (24/7 for production-down)
- Remote support via TeamViewer Tensor: machine ID is shown on the bottom-right of the HMI status bar.